

MATERIAL SAFETY DATA SHEET

Section 1. Chemical Product and Company Identification

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Product Name: Clethodim 2400g/L EC

Chemical Name of Active Ingredient (IUPAC): _

(±)-2-[(E)-1-[(E)-3-chloroallyloxyimino]propyl]-5-[2-(ethylthio)propyl]-3-hydroxy cyclohex-2-enone

Chemical Formula of Active Ingredient:

C₁₇H₂₆ClNO₃S

CAS/EPA/EU Registry Number of Active Ingredient:

99129-21-2 (CAS); 508 (CIPAC)

Section 2. Composition/Information on Ingredients

Component	CAS Number	Content (g/L)
Clethodim	99129-21-2	Min. 240
Inerts	Not applicable	Balance to 1 L

Section 3. Hazards Identification

CAUTION! Harmful if swallowed. May cause skin or eye irritation. Avoid contact with eyes, skin, or clothing. Do not breathe vapor or mist. Keep out of reach of children. Keep away from food, drink and animal feed.

Section 4. First Aid Measures

Contact: Flush with running water for at least 15 minutes while holding eyelids open to help flush out material. Get medical attention.

Skin Contact: Remove contaminated clothing and shoes immediately. Wash thoroughly with soap and water. Get medical attention.

Inhalation: Wearing proper respiratory protection before rescues. Immediately remove the affected victim from exposure to an area of fresh air. If not breathing, give artificial respiration; if breathe difficult, give oxygen. Get medical attention.

Ingestion: If conscious, give water or milk. Do NOT induce vomiting. Never give anything by mouth to an unconscious person. Get medical attention immediately.

Antidote: No special antidote. Treat symptomatically and supportively.

Section 5. Fire Fighting Measures

Fire and Explosion Hazards: Flash point: 64°C. This product can form combustible mixtures at temperatures at or above the flash point. It may produce a floating fire hazard in extreme fire conditions.

Firefighting Media: Use water, foam, carbon dioxide, or dry chemical to extinguish fires.

Firefighting Precaution: In the event of a fire, wear full protective clothing and NIOSH-approved self-contained breathing apparatus with full facepiece operated in the pressure demand or other positive pressure mode. Keep unnecessary people away. Dike area of fire to prevent material run-off. Decontaminate emergency personnel with soap and water before leaving the fire area. Avoid breathing dusts, vapors and fumes from burning materials. Control run-off water.

Section 6. Accidental Release Measures

In case materials are released, contact emergency response personnel. Keep unnecessary persons away. If emergency response personnel are unavailable, collect leaking and spilled liquid in sealable containers as far as possible. Absorb remaining liquid in sand or inert absorbent and remove to safe place. Dike large spills and transfer to an appropriate container for disposal. Do NOT wash away into sewer. Use suitable protective equipment (Section 8). Follow all fire prevention procedures (Section 5).

Section 7. Handling and Storage

Handling Precautions: Protect against physical damage. Use non-sparking type tools and equipment, including explosion proof ventilation. Do not contaminate lakes, streams, and ponds. When using, do not eat, drink, or smoke. Avoid contact with skin and eyes. Do not breathe spray. Wash hands before eating, drinking, chewing gum, using tobacco, or using the toilet. Remove contaminated clothing and wash thoroughly after handling.

Storage Precautions: Store in original container only in cool, dry, well-ventilated, secure area out of reach of children and animals. Do not contaminate food or use on household tanks.

Section 8. Exposure Controls/Personal Protection

Engineering Controls: Ventilation equipment should be explosion-resistant if explosive concentrations of material are present. Provide local exhaust or process enclosure ventilation system.

Eye/Face: To protect against accidental eye contact, goggles/face-shield should be worn.

Skin Protection: Rubber gloves should be worn. Wash thoroughly with soap and water after handling.

Respiratory Protection: Ensure good ventilation. For maximum protection, wear a supplied air, full-facepiece respirator, airlined hood, or full-facepiece self-contained breathing apparatus.

Section 9. Physical and Chemical Properties

Appearance yellowish to brown liquid

Odor Aromatic odor

Specific Gravity Approx. 1.0 (25°C)

Water Solubility Disperse in water

Section 10. Stability and Reactivity

Stability: Stable under the normal handling and storage conditions.

Incompatibilities: Avoid mixed with strong acid, strong bases, strong oxidizing agent or substances.

Hazard Decomposition: The substance decomposes on heating and producing toxic fumes including carbon monoxide, carbon dioxides, nitrogen oxides, sulfur oxides and hydrogen chloride.

Hazard Polymerization: Will not occur.

Section 11. Toxicological Information

Acute Oral Toxicity: LD₅₀ for rats >2000 mg/kg

Acute Dermal Toxicity: LD₅₀ for rabbits >4000 mg/kg

Acute Inhalation Toxicity: LC₅₀ (4 h) for rats >5.0 mg/l

Skin Irritation: Mildly irritating to skin of rabbits. Not a skin sensitizer.

Eye Irritation: Slightly irritating to eyes of rabbits.

Toxicity Class: III (WHO), Slightly hazardous.

Section 12. Ecological Information

Do not contaminate dams, waterways or sewers with this product. Referenced technical active ingredient ecological information list as below:

Birds Acute oral LD₅₀ for bobwhite quail >2000 mg/kg. Dietary LC₅₀ for mallard ducks >6000 mg/kg.

Fish LC₅₀ (96 h) for rainbow trout 67, bluegill sunfish >120 mg/l.

Bees LD₅₀ (contact) >100 µg/bee.

Environmental Fate:

Animals Metabolized to the sulfoxide and sulfone.

Plants Metabolized to the sulfoxide, sulfone and S-methyl sulfoxide.

Soil/Environment

Aerobic DT₅₀ in soil 1-3 d. K_d 0.08-1.6 (5 soil types).

Section 13. Disposal Considerations

Disposal of product would be treated, stored, transported, and disposed of according to the local waste regulation authority. Do not flush to surface water or sanitary sewer system.

Section 14. Transport Information

IMO Classification

Proper Shipping Name: Pesticide, Liquid,
Clethodim 240g/L EC

Class: 9

UN No.: 3082

Packaging Group
Marine Pollutant

Section 15. Regulator Information

European/International Regulation: Not regulated.

Section 16. Other Information

Disclaimer: The above information contained herein is given in good faith and to the best of our knowledge. However, no warranty is expressed or implied.